DOCKHUB · RISK REGISTER AND HAZARD ANALYSIS

DockHub risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First risk register, preliminary hazard analysis (stage P1) and first-pass design FMEA (stage P3) for option A at the kit 1.9.1 upgrade

DockHub risk register and hazard analysis

The top project risk is cost and sourcing: the parts come to USD 1,519 against a USD 1,200 target and the pack supply depends on the SwapCell project. The top hazard is a thermal runaway of a lithium pack in a bay on a public sidewalk, controlled by the steel compartment, the detector, the aerosol unit, the vent plenum and the rule that no pack is charged outside its window; the highest design FMEA scores are a runaway that spreads to a neighbouring bay (RPN 120), a pack that overheats on a hot day (RPN 72) and gas that collects in the plenum once the fans stop (RPN 60). Scores are 1 to 5. Risk is likelihood times impact. Method: STANDARDS section 23.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No co-design partner	Candidate partner (Los Deliveristas Unidos, DKH-DEC-001) not yet approached	Requirements stay desk assumptions; no TRL 4	3	5	Amish	Approach the partner before TRL 4
P2	Over the cost target	USD 1,519 parts against USD 1,200; pad USD 265	Prototype is not built within the target	4	3	Amish	Keep as a target; use a host slab where an engineer confirms it; review at supplier selection (DKH-DEC-001 O3)
P3	SwapCell interface changes	Pack project revises v0.3	Receptacle, cradle and CAN set change	2	4	SwapCell project	Build to v0.3; conflicts raised with the SwapCell project (DKH-DDR-001)
P4	Fire certification route unclear	FDNY guide covers charging cabinets, not swap stations by name (DKH-REG-001 G1)	Cabinet cannot be sited	3	5	Design lead	Ask FDNY early; UL 1487 route costed at TRL 4
P5	Charger power factor too low	Cheapest certified chargers lack PFC	17.2 A at 120 V, over a 15 A circuit	3	3	Design lead	Buy PFC chargers; to confirm at purchase (DKH-DEC-001 to-confirm item 1)
P6	Pilot site too hot	Summer highs well above 30 C	Charging pauses in the afternoon, fewer swaps	3	3	Design lead	Choose a site with highs near 30 C (DKH-DEC-001); revisit cooling for hot climates
P7	Pack supply and deposits	Station-owned pool not funded	No charged packs to swap	3	4	Partner	Confirm the ownership model in co-design (DKH-DEC-001)

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Fire from a pack thermal runaway	Cell defect, abuse, over-temperature, crushed pack	Burns, smoke, fire spreading to the street or a building	5	3	Steel compartment per bay; detector within 10 s and contactor in 1 s; condensed aerosol; vent plenum to a rear roof hood; doors locked during an alarm; pack refuses charge outside 0 to 45 C; propagation test planned	3
H2	Toxic or flammable gas at the door	Venting pack	Irritation, burns	4	2	Gas leaves by the plenum to the rear roof; door locked during alarm; fans stop on aerosol discharge (DKH-DEC-001 D1)	2
H3	Electric shock	120 or 230 V mains in the technical compartment; 54.6 V pack terminals	Shock, burns	5	2	RCBO in the grid unit; mains only behind the locked service door; blind-mate receptacle; INTERLOCK coding resistor; isolate before service	2
H4	Door pinch	Hand in the door gap as the steel door shuts	Crushed finger	3	3	Doors close by hand on piano hinges with an EPDM gasket, no powered closing; broken edges	2
H5	Cabinet falls over	30 m/s gust on the canopy; vandal pulling	Crush injury, damage	5	2	Four M12 anchors into a 1.1 t pad, equilibrium factor 1.12; canopy lowest point 2.17 m	3
H6	Strain from lifting a pack	2.85 kg lifted from a low bay	Back or wrist strain	2	3	Connector 0.73 m and handle top 1.105 m above the ground; pack far under lifting limits	1
H7	Burn from a hot pack	A pack taken just after charging	Minor burn	2	2	Pack temperature window; released at 80 % state of charge	1
H8	Trip or collision on the sidewalk	Plinth and canopy edges in a pedestrian way	Fall, cut, bump	3	3	1.0 x 0.5 m footprint; broken edges; canopy above 2.1 m; private property first	2
H9	Theft and vandalism	Packs are valuable; unattended	Loss, forced entry, damaged locks	3	4	1.5 mm steel, fail-secure locks, anchored plinth, tokens plus a returned pack	3
H10	Installer injury at the pad	About 1.1 t of concrete, drilling into concrete	Dust, crush, strain	4	2	Mask and eye protection; lifting aid; 7 day cure before anchoring (build plan safety stops)	2

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Bay compartment and liner	Contain a runaway	Heat soaks through to the next bay	Second pack goes into runaway	0.7 to 1 MJ into about 7 kg of liner; the gap is air	5	4	6	120	Propagation trial with plain and vermiculite floors (DKH-DEC-001); heat-soak bound 248 K (STEM S11)
F2	Roof fans and aerosol unit	Vent gas and suppress flame	Fans sweep out the aerosol	Flame not suppressed	Fans ran on alarm	4	2	4	32	Fixed by D1: fans stop on discharge; retention test in a mock-up (S13)
F3	Charger and pack	Charge within the window	Pack above 45 C on a hot day	Charging pauses; pack ageing	Ambient near 30 C plus 4.2 K cabinet rise plus up to 10.8 K pack rise	3	4	6	72	Controller pauses; site choice; STEM S9
F4	Detector and NTC	Detect over 60 C or smoke in 10 s	Sensor lags or is blocked	Late alarm, late contactor	Thermal lag; dust in the bay	5	3	4	60	Bench test of detection time at TRL 4 (S26); clean at service
F5	Door gasket and hood	Keep IP54	Water or dust enters	Corroded contacts, charger failure	Cut gasket, blocked rain lip, clogged filter	3	3	4	36	Enclosure test at TRL 4 (S27); filter change on the service list
F6	Solenoid lock and door switch	Hold the door; confirm the return	Lock jams shut or fails open	Pack trapped, or released without a return	Water, wear, failed switch	3	3	3	27	Service release from inside; switch checked by the controller; spares
F7	Receptacle and cradle	Blind-mate the pack	Misaligned contacts, damaged pins	Pack not detected, or arcing	Cradle tolerance, abuse	4	2	4	32	Guide faces on the cradle; contact check before charge
F8	Anchors and pad	Hold the station down	Anchor pulls out in a gust	Cabinet moves or falls	Cracked concrete, wrong torque; 3.8 kN needed	5	2	4	40	Pull test each anchor; confirm the maker's rating (DKH-DEC-001 to-confirm item 9)
F9	Controller and CAN	Run the bays	Controller hangs	Bays stop; locks stay secure	Firmware fault, power dip	3	3	3	27	Watchdog; bays stay locked and chargers hold a safe state; 24 h offline operation

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F10	Canopy and panel	Shade and make power	Post bolt loosens	Panel shifts, hits rider	Vibration	4	2	4	32	Locking nuts; inspect at service
F11	Plenum with fans stopped	Vent hot gas after an aerosol discharge	Unburnt gas collects and ignites	Pressure pulse, door or hood damaged, flame at the hood	Fans stopped by fix D1; only the 240 cm2 hood slot vents	5	2	6	60	Calculate gas volume and an explosion-venting area; hold-time alternative (DKH-DEC-001 O4, STEM S29)

Sources

- DKH-CAL-001 for the loads, temperatures, latencies and masses quoted; DKH-STM-001 for the claims behind F1 to F4; DKH-REG-001 for P4.
- Hazard and FMEA scores are the author's judgment on a 1 to 5 scale, not field data.